



Khulna University of Engineering & Technology

Department of Computer Science and Engineering

Final Project Report

DormMate

Smart roommate matching platform

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DBMS:	SQLite Database
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■ 1.Project Overview

DormMate is a modern web application that completely changes how university students search for and match with roommates. Typically, finding a suitable roommate involves scrolling through disjointed social media posts, handling endless group chats, and taking a chance on compatibility. DormMate simplifies and automates this entire process by introducing a structured Peer-to-Peer roommate finder system.

The core purpose of the platform is to create a secure, data-driven space where students can outline their lifestyle preference factors, review compatibility percentages, read AI-generated reasons for compatibility, and communicate safely to reach a mutual roommate agreement.

How it works:

- **Smart Matching:** The system implements an automated matching logic that evaluates user profiles. As soon as a student creates a profile and sets their preferences, the system calculates a compatibility score against all other students.
- **AI Compatibility Details:** To assist students in matching decisions, the platform integrates with the Gemini API to explain compatibility factors in natural, descriptive language.
- **Direct Communication:** Matched users can immediately chat privately within the portal to negotiate and finalize their roommate agreement.

■ 1.1Strategic Objectives:

- **Roommate Discovery:** build a clean, responsive portal where students can search for compatible roommates quickly.
- **Attribute Profiling:** To match students based on sleep patterns, study habits, noise tolerance, and cleanliness.
- **Secure Messaging:** To enable private chat communication for matched users to safely coordinate terms.
- **Data Integrity:** To manage database records, user preferences, and roommate request states safely using SQLite and Laravel.

■ 2.Core Features:

Authentication & Identity Management :

- **Secure Registration:** Collects essential student credentials, department details, roll numbers, and safely encrypted passwords.
- **Safe Login Sessions:** Manages state tokens to keep students logged in and clears session cookies on logout.
- **Account Control:** Separates standard students from administrative moderators to manage platform safety.

Profile & Preference Management :

- **Lifestyle Profiling:** Allows students to configure their department, batch, hall of residence, and bio.
- **Preference Settings:** Captures sleeping schedules, cleanliness standards, study habits, noise tolerance, and hobbies.

Algorithmic Matching Engine :

- **Compatibility Calculator:** Utilizes lifestyle attributes to generate a compatibility percentage for matches.
- **AI Explanations:** Integrates with the Gemini API to explain the reasoning behind match scores in natural language.
- **Top Match Vectors:** Automatically lists the most compatible roommates on the student's home dashboard.

Roommate Request Architecture :

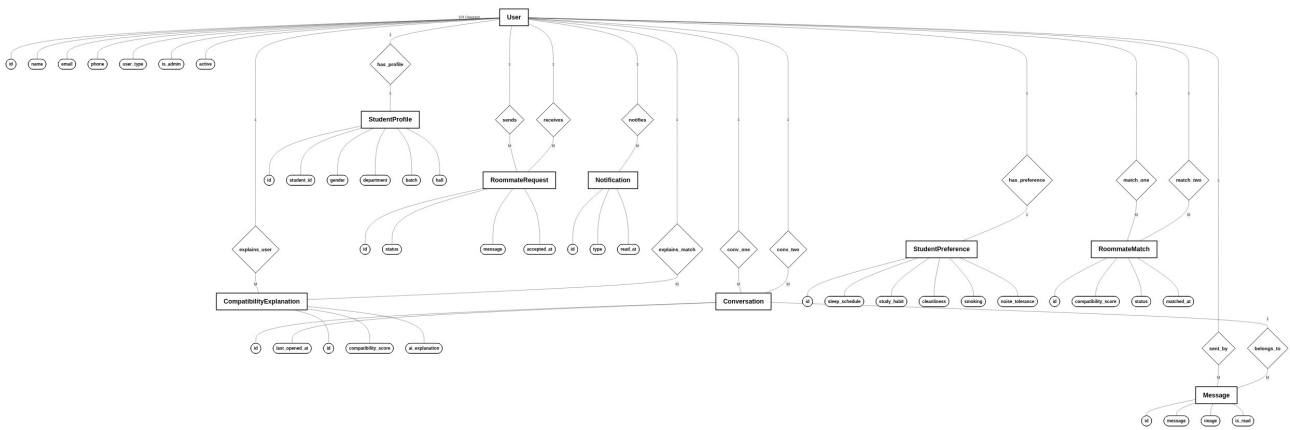
- **Request-Accept Flow:** Protects students by requiring a roommate request to be approved before final matching.
- **Mutual Agreement Rule:** Restricts students to a single active roommate to enforce 1-to-1 matching.
- **Private Chat Interface:** Creates a private messaging window automatically upon roommate request initiation.

2.1 Step-by-Step Workflow:

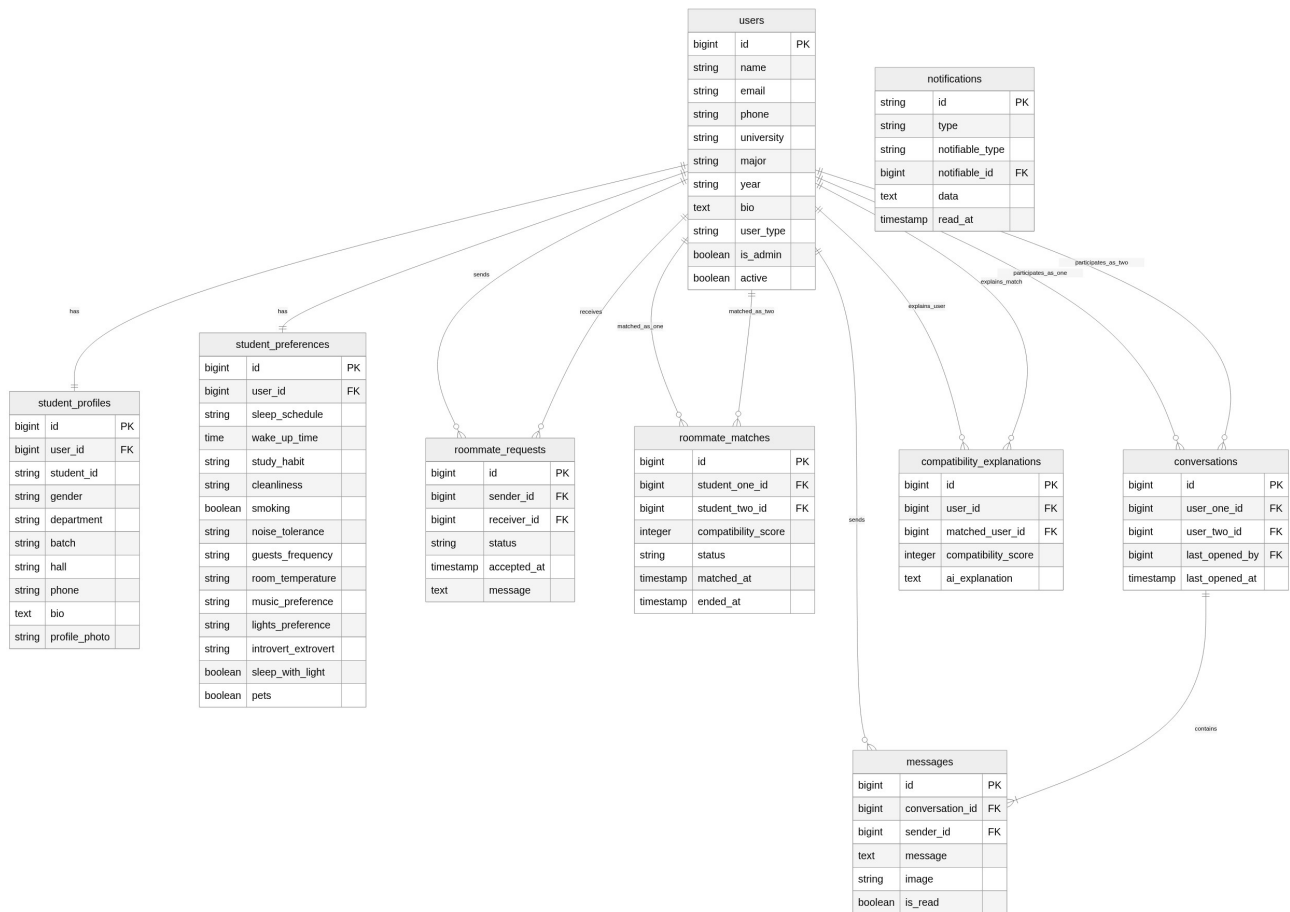
- **Step 1: Sign Up & Login:** Students register with university credentials and log in to the portal.
- **Step 2: Create Profile & Preferences:** Students provide gender, department, batch, and lifestyle preferences.
- **Step 3: Compatibility Matching:** The system calculates compatibility percentages and lists top matches.
- **Step 4: Review AI Explanations:** Students read the natural language compatibility explanation generated by AI.
- **Step 5: Send Request & Chat:** Students send a roommate request and chat privately to coordinate.
- **Step 6: Finalize Roommate Match:** Once a match is confirmed, the roommate status is locked, preventing multiple assignments.

2.2 TECHNOLOGY STACK:

Technology	Role in Project	What it Actually Does
Laravel 12	Backend Framework	Backend engine handling routing, database migrations, controllers, and validation.
SQLite Database	Database	Stores user credentials, student profiles, lifestyle preferences, and chat logs.
Bootstrap 5	UI Styling	CSS framework providing a clean, responsive layout for mobile and desktop screens.
HTML5, CSS3, & JS	Web Designing	Structures user forms, styles matching lists, and enables interactive chat.
Gemini API	AI Explanations	Generates natural language descriptions explaining roommate compatibility.
Eloquent ORM	Database Logic	Simplifies data models, relationships, and queries between Laravel and SQLite.



3.2 SCHEMA DIAGRAM :



4.1 DISCUSSION & CONCLUSION:

Discussion:

- **Data Models:** DormMate successfully maps student entities, profiles, preferences, and roommate requests into structured SQLite database tables.
- **Algorithmic Evaluation:** application automates compatibility calculations based on student preferences, providing students with structured matches.
- **AI-Assisted Decisions:** Integrating the Gemini API allows the portal to explain compatibility percentages in natural language, helping students make informed roommate selections.

Conclusion:

DormMate provides a secure, efficient, and user-friendly solution to the common student housing challenge of finding compatible roommates. By automating compatibility scores, explaining matches using AI, and providing a private communication chat, the portal ensures that student accommodation remains safe, compatible, and mutually agreed upon.

4.2 GITHUB REPOSITORY LINK:

GITHUB LINK: <https://github.com/lmtiaz2207092/DormMate>